Integrated Himalayan EWS — System Architecture

Binding monsoon flooding & landslides / ice–rock avalanche / GLOF / major earthquake onto the two systems Nepal already runs (working draft)

Drafted 2026-08-30 · Based on analysis of the 26 August 2026 Bhote Koshi–Trishuli flood · Not an official position

Do not build from zero. Connect the two systems that exist; add only the missing detection paths.

1 The two systems Nepal already has

- Rain **DHM rainfall & river "live watch": 280 stations**
  Web telemetry since 2010. The backbone of the rainfall track
- Rain **Threshold-based flood & landslide warning (since 2018)**
  272 hazard areas delineated; SMS is sent when upstream rainfall crosses threshold
- Rain **CBFEWS in 8 river systems · 3-day rainfall forecast**
  Koshi, Karnali, West Rapti and others. **Narayani is included — already present in this corridor**
- Rain **1 of 3 planned weather radars**
  Installed in the west in 2019. Central and east: not yet
- Seis **~40 broadband seismometers · ~36 accelerometers**
  DMG / NEMRC, plus 8 Univ. of Tokyo–JICA stations and 10 China CEA stations

For rainfall-driven hazards, the chain from detection to dissemination is already complete. What failed here was not the absence of a system but the fact that **the system presupposes rain.**

2 So why did nothing fire?

- Gap 1 **The rainfall track presupposes rainfall**
  No rain was involved in this event (confirmed by DHM). Neither the 280 stations nor the 272 area thresholds **could respond in principle**
- Gap 2 **The seismic track presupposes earthquakes**
  Associators are built on sharp arrivals. A slow-onset mass movement is discarded at association. **nst = 0** in this event's catalogue entries is the evidence
- Gap 3 **The two tracks are not wired together**
  Even with an M5.2 on the record, nothing carries it into the flood alerting path. **Both 08:37 and 11:45 reached the catalogue only after an analyst processed them — posted a day and a half later**
- Gap 4 **Mandates are split**
  Earthquakes = DMG / flood & landslide warning = DHM / response = NDRRMA

Each track works for its own hazard. Neither owned this hazard, and they were not connected to each other. That is what "integration" has to fix — and none of Gaps 1–4 is a shortage of hardware.

3 Hazard stocktake — what is caught, and by which path

HAZARD	DETECTION PATH	STATUS
Monsoon flooding & landslides Cumulatively the largest killer of all	**R** rainfall threshold + soil water index	existing operating
Ice–rock avalanche / rock collapse 2026 this event / 2015 Langtang	**D1** single-force source alert in 60–120 s	new
GLOF / landslide-dam breach 2016 Gongbatongsha / 2025 Rasuwa	**D2** fluvial tremor (1–10 Hz)	new
Loss of the instruments themselves Here, 3 upstream gauges fell silent below alert stage	**D3** missing data = alarm	new
Widespread coseismic slope failure 25,000 landslides in Gorkha 2015	**State modifier** lowers every trigger threshold	new

What this system cannot catch — stated explicitly

- **Snow avalanche** Detectable as a small-magnitude D1 class. Forecasting is impossible — no snowpack observation exists
- **Earthquake early warning itself** Nepal has no national EEW (demonstration stage only)
- **Air blast** Occurred at Langtang 2015 and Chamoli 2021. Damage reaches beyond the flow path

4 Six-layer architecture — what is reused, what is modified, what is added

L0 Baseline risk — sets where sensors go
- E **ICIMOD / UNDP glacial lake inventory, 47 lakes** — 28 highly susceptible in the Poiqu-Gyirong transboundary basins
- M **Static inventory → periodic SAR detection of new water** — the 2025 supraglacial lake formed in six months, on no list
- N **InSAR screening of unstable slopes** — Langtang Lirung also failed in 2015
- M **Exposure mapping** — settlements, roads, hydropower adits, trekking routes

L1 Sensing — rainfall + seismic
- E **DHM rainfall & river "live watch", 280 stations** — backbone of the rainfall path. Operational
- E **Weather radar, 1 of 3** — west only, installed 2019
- E **NSC broadband seismometers, ~40** — DMG / NEMRC
- E **36 accelerometers, 8 Univ. Tokyo-JICA** — China CEA adds 10 across the north
- M **DHM stage gauges, 6 in the corridor** — → 30. Trigger on rate of rise
- N **3-5 broadband + strong-motion per headwater basin** — on bedrock, above any credible flood level
- N **Co-located microbarometers** — earthquakes emit little infrasound - it separates mass movement
- N **Riverside tremor stations** — a non-contact gauge the flood cannot destroy
- M **GSMaP / GPM for transboundary rainfall** — no ground observation on the Tibetan side

L2 Telemetry
- E **Existing telemetry to Kathmandu** — cellular / microwave. Follows the valley, so the same event severs it
- N **Satellite IoT as a second path** — the alert path carries parameters, not waveforms. Assume total terrestrial loss in a major earthquake

L3 Processing — joining two systems
- M **Rainfall path: thresholds alert 272 delineated areas (since 2018)** — operational, but blind to transboundary rainfall and short on lead time
- E **Seismic path: associate impulsive arrivals to locate** — emergent mass-movement signals are discarded at association (nst = 0 here)
- M **R Rainfall-runoff forecasting, strengthened** — soil water index / snake curve
- N **D1 Single-force detection** — mass movement. Alert in 60-120 s
- N **D2 Fluvial tremor detection** — 1-10 Hz. Bedload impacts = flood
- N **D3 Loss-of-signal detection** — silence is evidence of damage
- N **Fusion, three confidence levels** — accepts both rainfall and non-rainfall paths
- N **A major earthquake is a state modifier, not a fifth trigger** — 2015 Gorkha M7.8 set off some **25,000 coseismic landslides** and the Langtang rock-ice avalanche (350+ dead). On detection, lower the R / D1 / D2 thresholds, auto-task satellites, and assume river blockage

L4 Decision
- N **Upper 20 km: sensor to siren, no human in the loop** — a ten-minute margin leaves no room for procedure
- M **Downstream: feed arrival-time forecasting → per-village ETA** — one shared outlet for both the rainfall and non-rainfall paths
- E **NDRRMA and DHM hold the authority to issue, but no body is mandated to carry a seismically detected mass movement into the flood alerting path**

L5 Dissemination
- E **CBFEWS, 8 river systems** — includes the Narayani - already present in this corridor
- M **SMS, radio, TV, telephone** — 679,295 sent. Extend the area to the hazard's
- N **Cell broadcast** — multilingual; reaches roaming foreign SIMs
- N **Underground works alarm** — matches the 933 missing tunnel workers
- N **QZSS DC Report** — receivability PoC required

Colour encodes status. **Grey = existing, used as is** / **red outline = existing, but did not function here** / **teal = existing, to be modified** / **blue = newly added**. The only permanent new installations are the three L1 items and the satellite IoT in L2. **Integration lives in L3** — the four triggers stay separate, and only the L4 arrival-time prediction and the L5 dissemination paths are shared. No single detector is bet on; each track can raise an alert independently. **Confidence has three steps** (one detector = advisory / two detectors agreeing = prepare to evacuate / downstream measurement = evacuate). **L0 is the peacetime layer**, deciding where stations go and what the thresholds are — a static glacial-lake inventory is not enough: the 2025 supraglacial lake appeared within six months.

5 Phasing — Phase 0 is the go / no-go

- Phase 0 **Retrospective analysis of existing archives** — 0–6 months · < USD 0.35 M · no hardware
  Mine ~40 stations' continuous waveforms backwards and label past events. **Training data, thresholds and false-alarm rate fall out together.**
- Phase 1 **Shadow-mode operation** — 6–18 months · < USD 0.7 M
  Run the detector alongside the existing processing chain. Issue no warnings; **measure the operational false-alarm rate.**
- Phase 2 **Implement in a pilot basin** — 18–30 months · USD 1.4–2.8 M
  Fill station gaps in the Bhote Koshi–Trishuli; add infrasound and riverbank tremor sites. Automatic sirens within 20 km upstream; arrival-time prediction downstream.
- Phase 3 **Roll out to other basins · transboundary integration** — USD 1.4–2.1 M per basin

If Phase 0 shows events cannot be caught with adequate margin, nothing further should be funded. It costs almost nothing and governs a multi-million-dollar decision.

6 Three things harder than the technology

1. **Assign the mandate**
   Widen DHM's flood-warning remit to accept non-hydrological triggers, or stand up a joint DMG–DHM–NDRRMA cell. **Without it, detection has no route to a warning.**
2. **Agree the false-alarm budget before go-live**
   Settle a line with communities — "three false alarms a year are acceptable if no river-threatening event is missed" — and rehearse it in drills. **A system whose first false alarm comes as a surprise is switched off within a year.** A political decision, not a technical one
3. **Transboundary**
   The source sits on the Tibetan side, but **seismic waves cross borders**. Stations inside Nepal can detect it, so the system works without waiting for a data-sharing agreement — the political strength of this design

Reference design: the Illgraben debris-flow warning system in Switzerland has run **fully automatically since 2007**, with no human between check-dam geophones and the alarm. "Wire the sensor straight to the siren" has worked in the field for nearly two decades.

7 What this system cannot do

- **It does not predict. It begins once the mountain moves**
  The Chamoli 2021 precursor (2.5 h ahead) was found retrospectively. **A research track — never a design assumption.**
- **Tremor amplitude is not discharge**
  A proxy for sediment flux; calibration is site-specific. A small GLOF can hide inside a monsoon rise. The discriminants are **rate of change and spectral shape**.
- **Below threshold, nothing is seen**
  The smallest volume that can block a channel is unknown. The threshold is a choice of which risk is accepted — Phase 0 turns it into a number.
- **It does not replace stage gauges**
  Gauges give calibration; seismometers give survivability. **Complementary, not substitutes**. Leaning on either alone repeats the failure in another form.

Legend (colour coding of the architecture)

E = existing, used as is | E = existing, did not function here | M = existing, to be modified | N = newly added

New permanent hardware is limited to **the three L1 items and the L2 satellite IoT**. The rest is **software and institutions**.
Integration lives in L3 — triggers stay separate; only decision and dissemination are bound together.

What Japan can bring

- **NIED**: Hi-net / V-net; operational automatic lahar detection
- **Univ. of Tokyo & JICA**: 8 stations already in Nepal — a base to extend
- **MLIT Hokkaido Bureau**: the Tokachidake sensor-to-siren direct-wiring model
- **PWRI / ICHARM**: turning detection into arrival times and evacuation lead time
- **JMA & JAXA**: landslide warning information (snake curve); GSMaP upstream rainfall
- **JAXA · Sentinel Asia · ADRC**: satellite monitoring of the upstream across the border

Alert regardless of mechanism.

The system must not try to decide "rock or ice". The only question it answers is binary: has a large mass movement occurred upstream? Identifying the mechanism takes days — four days on, views still diverge.